JAN-M-621

14 JULY 1948

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

MAGNESIUM FLUORIDE
(Optical Use)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

U. S. Army
Frankford Arsenal
Tentative Specification
FXS-952 (Rev. 2)
17 June 1945

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

A-1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. ARMY SPECIFICATION
100-2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION
General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATION
RR-S-366—Sieves; Standard, Testing.

A-2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION
Navy Shipment Marking Handbook.[2]

B. CLASSES

B-1. *Classes.*—Magnesium fluoride shall be of the following classes, as specified in the contract or order (see par. H-2):
Class a—For use as an abrasive.
Class b—For use in producing films on optical glass.

C. MATERIAL AND WORKMANSHIP

C-1. See section E.

D. GENERAL REQUIREMENTS

D-1. See section E.

E. DETAIL REQUIREMENTS

E-1. *Color.*—Magnesium fluoride shall be white.

E-2. *Chemical requirements.*—

E-2a. *Class a.*—Magnesium fluoride for use as an abrasive shall be 99.8 percent pure. (See par. F-5a.)

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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E–2b. *Class b.*—Magnesium fluoride for use in producing films on optical glass shall be 99.8 percent pure.

E–2b(1). *Chloride content (Class b).*—Magnesium fluoride shall not contain more than 0.001 percent chlorides (calculated as $MgCl_2$).

E–3. *Granulation.*—

E–3a. *Class a.*—100 percent shall pass through a No. 100 sieve.

E–3b. *Class b.*—90 percent shall pass through a No. 20 sieve and not more than 25 percent shall pass through a No. 200 sieve.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Size of lot.*—Maximum, 200 pounds. Unless otherwise specified in the contract or order, the lot shall consist of material from not more than one manufacturing batch of magnesium fluoride. (See par. H–3.)

F–2. *Sampling.*—Select 10 percent but not more than 10 in number of the containers comprising the lot. Remove a sample from each selected container and label each primary sample so that the container from which it was taken can be identified. Make a composite sample of about 120 grams. Mix this thoroughly. Make all acceptance tests on the composite sample. Place all samples in glass-stoppered bottles or mason jars and label each container giving the name of the material, manufacturer, plant, contract or order number and the number of pounds in the lot. The primary samples will be held by the inspector for possible future examination, should the composite sample fail to meet the requirements.

F–3. *Inspection.*—Inspection shall be at the point of delivery unless otherwise specified in the contract or purchase order.

F–4. *Color.*—Compliance with the requirements of paragraph E–1 shall be determined by visual inspection.

F–5. *Tests.*—

F–5a. *Chemical analysis.*—A sample shall be subjected to chemical analysis to determine compliance with the requirements of paragraph E–2. Analytical methods used shall be satisfactory to the bureau or service concerned. Analysis of Class a magnesium fluoride may be waived at the option of the bureau or service concerned.

F–5b. *Chloride content.*—Finely pulverize about 5 grams of the composite sample in a clean mortar. Weigh 4.0 grams of the pulverized sample into a test tube and add 5 ml of distilled water which has been previously checked to give no cloudy appearance with silver nitrate solution. Boil the mixture in the test tube for 5 to 10 minutes, taking care not to lose any water by bumping. Stopper the test tube and allow the solution to stand for 15 hours or more. At the end of this period, add 3 to 5 drops of 5-percent silver nitrate solution and 3 to 5 drops of reagent-grade nitric acid to the clear supernatant liquid. Do not shake or stir. At the same time pour about 5 ml of a standard solution containing

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10 parts per million of sodium chloride (made by dissolving 1 gram of sodium chloride in 100 ml of distilled water and diluting 1 ml of this solution to 1000 ml with distilled water) into another clean test tube and add a few drops of the silver nitrate solution. A slight cloudy precipitate of silver chloride will appear which requires about 5 minutes to reach maximum turbidity. If the turbidity in the sample at the end of 5 minutes is greater than that in the standard, it shall be cause for rejection of the lot.

F–5c. *Particle size.*—Weigh approximately 100 grams of the sample and transfer to the sieve or nest of sieves and receiving pan, as specified in paragraph E–3. Sieves shall comply with Federal Specification RR–S–366. Cover and shake for 10 minutes by means of a single-eccentric type mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Report the percentage passing through each sieve.

F–6. *Rejection and resubmission.*—If the composite sample fails to comply with any of the specified requirements the entire lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on each container in the lot at no expense to the Government. The contractor may then remove defective portions of the lot and resubmit the lot for acceptance. (See par. H–4.)

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging.*—Unless otherwise specified in the contract or order (see par. H–2) magnesium fluoride shall be packaged in 4-ounce standard square bottles with narrow neck. Screw-caps shall be of molded plastic with cork insert faced with aluminum foil to provide a non-leakable seal. Empty bottle weight should not exceed 5 ounces.

G–2. *Packing.*—Unless otherwise specified, magnesium fluoride shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carrier, for safe transportation at the lowest rate, to the point of delivery.

G–3. *Marking.*—

G–3a. *Packages.*—Each bottle or other package shall be labeled to show name of contents, the specification number and the class.

G–3b. *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the material, the class, and quantity contained therein as specified in the contract or order, the name of the contractor, and the number of the contract or order. In addition, shipments for the Army shall be marked in accordance with U. S. Army Specification 100–2; for the Navy, in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Use.*—Class a magnesium fluoride as covered by this specification is intended for use in the testing of reflection-reducing films on optical glass. Class b magnesium fluoride is intended for use as a raw material in the production of reflection-reducing films.

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H–2. *Ordering information.*—Requests, requisitions, schedules, and contracts or orders should specify the following:

(a) Title, number and date of this specification.
(b) Class required. (See par. B–1.)
(c) Type of package if other than 4 ounce bottle is desired. (See par. G–1.)

H–3. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

H–4. *Salvage.*—Magnesium fluoride which is rejected because of excessive chloride content may be salvaged by starting with the defective material and again carrying through the chemical procedure of its manufacture.

H–5. Copies of National Military Establishment, Joint Army-Navy, and Federal specifications (required for Army and Air Force purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the Department of the Army and the Department of the Air Force will obtain copies of National Military Establishment, Joint Army-Navy, U. S. Army, and Federal specifications through established Departmental channels. Both the title and identifying symbol number should be stipulated when requesting copies.

H–6. Copies of National Military Establishment, Joint Army-Navy, and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Department of the Navy, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol number should be stipulated when requesting copies.

H–7. Copies of this National Military Establishment specification (required for Army purchases) may be obtained from the Office, Chief of Ordnance, Department of the Army, Washington 25, D. C.

Notice.—When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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